UNIVERSITI TEKNOLOGI MARA
FINAL EXAMINATION

COURSE	INTRODUCTION TO INTERACTIVE MULTIMEDIA
COURSE CODE	CSC413
EXAMINATION	NOVEMBER 2026
TIME	3 HOURS

INSTRUCTIONS TO CANDIDATES

1. This question paper consists of the questions printed in this booklet.
2. Answer ALL questions in the Answer Booklet. Start each answer on a new page.
3. Do not bring any material into the examination room unless permission is given by the invigilator.
4. Please check to make sure that this examination pack consists of:
   i) the Question Paper
   ii) an Answer Booklet - provided by the Faculty
5. Answer ALL questions in English.

DO NOT TURN THIS PAGE UNTIL YOU ARE TOLD TO DO SO

This examination paper consists of 5 printed pages.

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PART A (15 MARKS)

State TRUE (T) or FALSE (F) for the following statements.

1. A plotter is an input device.
2. Sampling converts a continuous sound wave into digital values.
3. Anti-aliasing makes the edges of text look more jagged.
4. The amplitude of a sound wave determines its loudness.
5. Streaming allows media to begin playing before the whole file is downloaded.
6. A linear presentation allows users to choose their own path.
7. Multimedia combines text, graphics, audio, video and animation.
8. Kerning adjusts the space between individual character pairs.
9. Authoring tools help developers organise and edit multimedia elements.
10. Interactivity level is irrelevant when choosing an authoring tool.
11. CMYK is used for printed output.
12. RAM is a type of permanent storage.
13. Path animation moves an object along a predefined route.
14. Hertz measures sound frequency.
15. Frame rate is the number of video frames displayed per second.

PART B (15 MARKS)

Select the most appropriate answer from Table A for each of the following questions.

Table A. Matching options

A. Multimedia	B. Linear multimedia	C. Tweening
D. Virtual reality	E. Scanner	F. Frame rate
G. Chroma key	H. Keyframe	I. Bitmap
J. Cel animation	K. Card-based tool	L. Vector
M. Copyright	N. Serif	O. Storyboard

1. exclusive rights of a creator
2. input device that digitises documents
3. important frame in an animation
4. font with small strokes at character ends
5. image made of pixels
6. generating in-between frames
7. hand-drawn frame-by-frame animation
8. integration of text, graphics, audio, video and animation
9. presentation without user control of sequence
10. visual sequence plan
11. frames displayed per second
12. image made of mathematical objects
13. simulated environment
14. authoring tool organised like pages or cards
15. replacing a colour background

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PART C (30 MARKS)

Choose the correct answer for each question.

1. Hypermedia is best described as:
   A. A single video file
   B. A printed manual
   C. Multimedia with hyperlinks
   D. Text-only media

2. CMYK is used mainly for:
   A. Animation
   B. Sound mixing
   C. Printed output
   D. Screen display

3. Card-based authoring tools are suited to:
   A. Live video only
   B. 3D modelling only
   C. Messages with a beginning and end
   D. Audio editing only

4. Which image type scales without losing quality?
   A. Vector
   B. Photograph
   C. Scan
   D. Bitmap

5. Which best defines multimedia?
   A. A printed book
   B. Text, graphics, audio, video and animation
   C. Text only
   D. A single image

6. Which is an output device?
   A. Mouse
   B. Scanner
   C. Plotter
   D. Keyboard

7. Amplitude of a sound wave relates to:
   A. Frame rate
   B. Pitch
   C. Colour
   D. Loudness

8. A presentation where the user controls the sequence is called:
   A. Non-linear
   B. Printed
   C. Linear
   D. Static

9. A higher sampling rate generally:
   A. Changes colour
   B. Reduces accuracy
   C. Improves accuracy
   D. Removes audio

10. Path animation moves an object:
   A. Only vertically
   B. Without motion
   C. Along a predefined route
   D. Randomly

11. A browser is used to:
   A. Scan images
   B. Display web pages
   C. Print documents only
   D. Edit audio

12. Which is an authoring tool category?
   A. JPEG
   B. Card-based
   C. RGB
   D. CMYK

13. Morphing changes:
   A. Audio pitch
   B. Font size
   C. File format
   D. One shape into another

14. Which is a benefit of multimedia in education?
A. It uses only text
B. It replaces all teachers
C. It combines visuals and sound
D. It removes all interaction

15. Which colour model is subtractive?
A. CMYK
B. RGB
C. HSV
D. HSL

16. Which is an example of linear multimedia?
A. An interactive kiosk
B. A website with menus
C. A cinema presentation
D. A game with choices

17. A hyperlink links to:
A. Only images
B. Only audio
C. Only text
D. Another resource

18. RGB is used mainly for:
A. Screen display
B. Printed output
C. Video editing only
D. Audio recording

19. An authoring tool is used to:
A. Only record audio
B. Assemble multimedia elements
C. Only browse the web
D. Only print text

20. Interactivity in multimedia means:
A. There is no navigation
B. The user can control content and flow
C. The system plays automatically
D. Only text is shown

21. Sampling converts:
A. Analog sound to digital values
B. Text to speech
C. Images to vectors
D. Digital video to text

22. A graphics tablet is used for:
A. Web browsing
B. Drawing input
C. Sound output
D. Printing

23. A wireframe is:
A. A final video
B. A sound file
C. A low-fidelity layout sketch
D. A codec

24. Streaming allows media to:
A. Play while downloading
B. Remove compression
C. Require full download first
D. Only play offline

25. Chroma key is used to:
A. Add text
B. Scan images
C. Compress audio
D. Replace a colour background

26. HTML is used to:
A. Compress video
B. Model 3D objects
C. Structure web content
D. Store audio samples

27. Keyframes are:
A. Audio samples
B. Colour profiles
C. Important frames in an animation
D. Web links

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28. Interlacing:
A. Draws every line at once
B. Adds sound
C. Draws alternate lines of a frame
D. Removes colour

29. Which format is commonly used for bitmap images?
A. MPEG
B. HTML
C. PNG
D. MIDI

30. Plug and play means:
A. Automatic device recognition
B. Wireless only
C. No power needed
D. Manual driver installation only

PART D (40 MARKS)

QUESTION 1

A museum wants a website that presents its collection and lets visitors explore related exhibits.

a) State TWO differences between linear and non-linear navigation.

(2 marks)

b) Describe TWO roles of hyperlinks in the website.

(4 marks)

c) Explain how streaming and compression affect multimedia delivery on the web.

(4 marks)

QUESTION 2

A product catalogue must show sharp images on screen and in print.

a) State TWO differences between bitmap and vector images.

(3 marks)

b) Describe TWO factors that affect image quality.

(3 marks)

c) Explain how colour model and resolution affect screen and print output.

(4 marks)

QUESTION 3

Text is an essential element in a multimedia project. Its shape, size, design, colour, arrangement and linking all affect the message.

a) State TWO reasons sans serif is mostly used for screen display and serif is used for printed material.

(2 marks)

b) Describe TWO roles of hypertext in multimedia.

(3 marks)

c) Explain how text can have a massive impact on an informative multimedia project in the link and HTML context.

(5 marks)

QUESTION 4

You are required to animate a company logo. The logo depicts a planet orbiting the sun.

a) State TWO animation file formats suitable for this project.

(3 marks)

b) Describe the steps of cel animation and ONE advantage of using it.

(3 marks)

c) Differentiate 2-D and 2.5-D animation with ONE example for each.

(4 marks)

END OF QUESTION PAPER